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Anti-hyperuricemic and nephroprotective effects of *Smilax china* L.

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ABSTRACT

Ethnopharmacological relevance: *Smilax china* L., popularly known as "Jin Gang Ten", has been widely used as a traditional herbal medicine for the treatment of gout, rheumatoid arthritis and other diseases for a long time in China.

Aim of study: The present study was carried out to investigate the effect of *Smilax china* L. on hyperuricemia and renal dysfunction in induced hyperuricemic animals.

Materials and methods: Five fractions (petroleum ether, chloroform, ethyl acetate, n-butanol and residual ethanol fraction) of *Smilax china* L. were orally administered to potassium oxonate-induced hyperuricemic mice for three days. The xanthine oxidase inhibitory activities and modes of action of nine compounds isolated from ethyl acetate fraction (EAF) were then examined *in vitro*. Finally, different dosages of EAF were administered to 10% fructose-induced hyperuricemic rats.

Results: EAF (250 mg/kg) exhibited stronger anti-hyperuricemic activity in hyperuricemic mice compared with the other four fractions. Caffeic acid, resveratrol, rutin and oxyresveratrol isolated from EAF showed different inhibitory activities on xanthine oxidase *in vitro*, with the IC_{50} values of 42.60, 37.53, 42.20 and 40.69 μM, respectively, and exhibited competitive or mixed inhibitory actions. Moreover, EAF (125, 250 and 500 mg/kg) markedly reversed the serum uric acid level ($p < 0.05$, $p < 0.01$ and $p < 0.001$, respectively), fractional excretion of urate ($p < 0.05$, $p < 0.01$ and $p < 0.01$, respectively) and blood urea nitrogen ($p < 0.05$, $p < 0.01$ and $p < 0.01$, respectively) to their normal states, and prevented the renal damage against tubulointerstitial pathologies in hyperuricemic rats.

Conclusion: These findings show that *Smilax china* L. exhibits anti-hyperuricemic and nephroprotective activity in hyperuricemic animals.

1. Introduction

Gout is usually characterized by recurrent attacks of acute inflammatory arthritis with a red, tender, hot, and swollen joint. It is caused by elevated levels of uric acid in the blood which crystallize and are deposited in joints, tendons, and surrounding tissues. Uric acid, the end product of purine metabolism, is created when xanthine oxidase (XOD) catalyzes the oxidation of hypoxanthine and xanthine in mammal. Renal insufficient excretion of uric acid is the primary cause of hyperuricemia in about 90% of cases, while overproduction is the cause in less than 10% (Wortmann, 2002). Impaired excretion of uric acid is mostly induced by urate transporter abnormality in the proximal kidney tubule. While production of uric acid is elevated due to more active XOD and high intake of dietary purine (Anzai et al., 2005; Caulfield et al., 2008).

Increasing clinical reports have shown that hyperuricemia associated with an increasing risk of not only gout, but also chronic nephritis, renal dysfunction, as well as metabolic syndromes (Iseki et al., 2001, 2004; Ishizaka et al., 2005; Yoo et al., 2005; Cirillo et al., 2006; Zhou et al., 2006; Weiner et al., 2008). Nowadays, a number of anti-hyperuricemic agents including uricosuric agents and XOD inhibitors have been available in the market (Schlesinger, 2004). The associated adverse reactions such as gastrointestinal irritation, bone marrow suppression, renal toxicity, hypersensitivity syndromes and so on, however, always limit their clinical uses (Horiuchi et al., 2000; Hammer et al., 2001; Terkeltaub, 2003). Moreover, patients have to use allopurinol which has an higher rate of hypersensitivity reactions when uricosuric drugs lose their efficacy in the case of concurrent renal insufficiency. Therefore, it is

Abbreviations: BUN, blood urea nitrogen; CF, chloroform fraction; DMSO, dimethyl sulphoxide; EAF, ethyl acetate fraction; FEUA, fractional excretion of urate; HPLC, high-performance liquid chromatographic; NF, n-butanol fraction; PEF, petroleum ether fraction; REF, residual ethanol mother solution fraction; Scr, serum creatinine; Sur, serum uric acid; Ucr, urinary creatinine; Uur, urinary uric acid; XOD, xanthine oxidase.

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very urgent and important to search for better anti-hyperuricemic medicines.

Smilax china L., popularly known as "Jin Gang Ten" in China, belongs to the Liliaceae family. It is prescribed to treat gout and rheumatoid arthritis by dispelling wind-evil and eliminating dampness according to the theory of Chinese medicine (State Administration of Traditional Chinese Medicine of People's Republic of China, 1999; Chen et al., 2008; State Pharmacopoeia Commission of People's Republic of China, 2010). It has also been used for syphilis, acute bacillary dysentery, tumor (Wu et al., 2010) and inflammation (Shu et al., 2006) for more than 1000 years and virtually no toxicity was reported. Stilbenes, flavonoids and steroidal saponins are reported as the main active components of *Smilax china* L. (Ruan et al., 2002; Xu et al., 2008). Although there are many reports of biological activities on gout, it is still little known about the anti-hyperuricemic mechanisms and active ingredients of *Smilax china* L. Therefore, the research was cried out to study the anti-hyperuricemic mechanisms and active components of *Smilax china* L.

Anti-hyperuricemic fractions of *Smilax china* L. were investigated in potassium oxonate-treated mice through examining serum uric acid (Sur) level. Subsequently, compounds were isolated from the anti-hyperuricemic fraction, and their XOD inhibitory activities were assayed *in vitro*. Finally, the anti-hyperuricemic mechanism and nephroprotective effect of ethyl acetate fraction (EAF) were also investigated through evaluating the indexes of fractional excretion of urate (FEUA), blood urea nitrogen (BUN), XOD and renal tubulointerstitial pathological changes in fructose-induced hyperuricemic rats. The final results provided a pharmacological basis on hypouricemia and renal protection of *Smilax china* L.

2. Materials and methods

2.1. Plant materials

The raw materials of the rhizome of *Smilax china* L. were collected in October 2008 in Anhui province, China, provided by Simcere Drugstore in Nanjing, China. The materials were identified by Professor Ping Li, School of Traditional Chinese Medicine of China Pharmaceutical University. A voucher specimen was deposited under the number no. 25441 in the herbarium of China Pharmaceutical University. All the materials were dried at the room temperature to constant weight.

2.2. Reagents and drugs

All chemicals were of analytical grade. Petroleum ether, chloroform, ethyl acetate and n-butanol were purchased from Nanjing Hanbang Chemical Reagent Company (Nanjing, PR China). Allopurinol, xanthine and XOD were bought from Sigma Chemicals (St. Louis, MO, USA). Potassium oxonate and fructose were purchased from Aldrich Chemical Company, Inc. Assay kits of XOD, uric acid, creatinine and BUN were obtained from Jiancheng Biotech (Nanjing, PR China).

2.3. Animals

ICR mice (male, 18–22 g) and Sprague–Dawley rats (male, 230–270 g) were purchased from the Laboratory Animal Center of Suzhou University (Suzhou, China) and were housed one week to adapt to their environment before used for experiments. All the animals were maintained on standard laboratory conditions of temperature $25 \pm 1\,^{\circ}C$ and a 12-h light/12-h dark cycle with free access to food and water for the duration of the study. All the experiments and animal care were performed strictly in accordance with the Provision and General Recommendation of Chinese Experimental Animals Administration Legislation and were approved by the Science and Technology Department of Jiangsu Province.

2.4. Anti-hyperuricemic effect of Smilax china L. fractions in potassium oxonate-induced hyperuricemic mice

2.4.1. Preparation of plant extracts

The dried powder of *Smilax china* L. rhizome (10 kg, 40-mesh) was percolated with 80% EtOH (1× 100 L) for six days at room temperature. The pooled extracts upon solvent were concentrated under partial vacuum to obtain 1200 g of ruddy residue (12%, w/w). The residue was extracted successively with equal volumes (9.6 L, for three times) of petroleum ether, chloroform, ethyl acetate and n-butanol. Each sub-fraction was then concentrated under reduced pressure to obtain petroleum ether fraction (PEF, 96 g, 8%, w/w), chloroform fraction (CF, 192 g, 16%, w/w), EAF (300 g, 25%, w/w), n-butanol fraction (NF, 480 g, 40%, w/w) and residual ethanol mother solution fraction (REF, 132 g, 11%, w/w).

2.4.2. Mice model of hyperuricemia and drug administration

All the mice were divided at random into eight groups of ten mice each. Hyperuricemic mice model induced by potassium oxonate (uricase inhibitor) was used to study drug action (Stavric et al., 1975; Zhu et al., 2004). Briefly, PEF (80 mg/kg), CF (160 mg/kg), EAF (250 mg/kg), NF (400 mg/kg), REF (110 mg/kg) and allopurinol (10 mg/kg) were dissolved in 0.3% CMC–Na aqueous solution, respectively. The dosage of each fraction corresponds to 8.33 g crude drug/kg. All drugs were given orally once daily at 8:00–9:00 a.m. for three consecutive days. Mice were intraperitoneally injected with potassium oxonate (250 mg/kg) 1 h before the final drug administration to increase the Sur level. Food, but not water, was withdrawn from the animals 1.5 h prior to drug administration.

2.4.3. Sample collection and measurements

Blood samples were collected from mice by tail vein bleeding 1 h after the final drug administration on the third day. The blood was allowed to clot for approximately 1 h at room temperature and then centrifuged at 2500 × g for 10 min to obtain the serum. The serum was stored at $-20\,^{\circ}C$ until use. The Sur level was measured using standard diagnostic kits. Each assay was performed in triplicate.

2.5. Isolation and identification of compounds from EAF

EAF was subjected to repeated chromatography on silica gel column and eluted with petroleum ether–acetone or CH_2Cl_2–MeOH gradient solvent system. Further purification was performed by using Sephadex LH-20 chromatography, preparing TLC and other methods. The structure of compound was confirmed by spectroscopic methods (UV, IR, MS, ^{1}H NMR and ^{13}C NMR).

2.6. Analysis of EAF by high-performance liquid chromatographic (HPLC)

The multi-components of the EAF of *Smilax china* L. were characterized by HPLC. The samples were analyzed using Aichrombond-1 C_{18} column (250 mm × 4.6 mm, 5 μm) with the detector wavelength set at 300 nm, and the mobile phase consisted of water containing methanol (A) and 0.1% (v/v) H_3PO_4 (B). A gradient program was used as follows: 0–30 min, 6–15% A; 30–60 min, 15–20% A; 60–100 min, 20–35% A. The flow rate was 1.0 mL/min.

2.7. Assay of XOD activity and inhibitory modes of action in vitro

2.7.1. Assay of XOD activity

The XOD activities with xanthine as the substrate were measured with spectrophotometer at 295 nm using the method reported previously (Yu et al., 2006) with the following modifications. The enzyme assay was initiated by the addition of 100 μL of xanthine (1 mM) to 900 μL of assay buffer containing 50 mM potassium phosphate buffer (pH 7.5), and 70 mU/mL XOD, with or without the tested samples. The tested samples solution were dissolved in dimethyl sulphoxide (DMSO) and subsequently diluted with phosphate buffer (pH 7.5) to a final concentration containing less than 1% DMSO (v/v). Allopurinol was used as a positive control at a final concentration of 10 μM in the assay mixture. The IC_{50} values of the samples were calculated from regression lines of a plot of the percentage of inhibition on XOD activity versus the concentrations of the samples.

2.7.2. Assay of XOD inhibitory modes of action

Enzyme kinetics was carried out in the absence and presence of the tested samples with varying concentrations of xanthine (0, 2, 3, 4, 7, 8, 13, 20 μM) as the substrate, where tested samples were at various concentrations (12.5–100 μM) and xanthine at a certain concentration, using the XOD assay methodology. The plots were drawn by the reciprocal of reaction velocity (v^{-1}) versus the concentrations of the substrate (1/[xanthine]). The inhibitory mode of action was determined on the basis of visual in section of the Lineweaver–Burk and Akaike information criterion. The K_i values were determined by fitting the enzyme activity substrate concentration data at various inhibitor concentrations to the equations for competitive inhibition, noncompetitive inhibition, and mixed-type inhibition (Origin version 6.1).

2.8. Anti-hyperuricemic and nephroprotective effects

2.8.1. Rat's model of hyperuricemia and drug administration

Rats were given drinking water (vehicle control rats) or 10% of fructose in drinking water (fructose-fed rats) with standard chow for eight weeks in plastic cages (Miatello et al., 2005). Fresh drinking water was replaced every two days. After four-week fructose feeding, fructose-fed rats were further divided into matched subgroups. Different group animals (8 rats in every group) were administered daily with water (1 mL/kg, as a vehicle control group), allopurinol (4 mg/kg, as a positive control group), EAF (125, 250 and 500 mg/kg), respectively. All drugs were given orally once daily at 9:00–11:00 a.m. for the subsequent four weeks.

2.8.2. Sample collection

Whole blood samples were collected from rats by cardiac puncture 1 h after the final drug administration. The blood was allowed to clot for approximately 1 h at room temperature and then centrifuged at 2500 × g for 10 min to obtain the serum. The serum was stored at −20 °C until use. The livers were also dissected quickly on the ice and homogenized in 9 volumes of 80 mM sodium pyrophosphate buffer (pH 7.4). The homogenate was then centrifuged at 3000 × g for 10 min and the supernatant was used for XOD assay. The kidneys were removed immediately and fixed in 10% formalin and processed in paraffin for subsequent histological assessment. Prior to cardiac puncture, rats were placed in metabolic cages for collection of 24 h urine, the volume of which was recorded for each group. Urine samples were centrifuged at 2000 × g for 10 min to remove the particulate contaminants and supernatant was used for analysis. All the samples were frozen at −80 °C until assay.

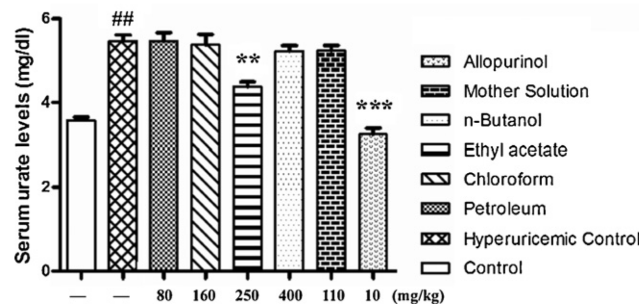

Fig. 1. Anti-hyperuricemic effects of various solvent partitioned fractions in potassium oxonate-induced hyperuricemic mice. Experiments were performed as described in Section 2. The samples were analyzed for Sur level. Data were expressed as mean ± S.E.M. of ten mice. Statistical analyses were performed by Student's *t*-test. ##$p < 0.01$ versus the control mice; *$p < 0.05$; **$p < 0.01$ and ***$p < 0.001$ versus the potassium oxonate-induced hyperuricemic mice.

2.8.3. Measurement of uric acid, creatinine, BUN and XOD activity

Sur, urinary uric acid (Uur), serum/urinary creatinine (Scr/Ucr) and BUN levels and hepatic XOD activity were measured using standard diagnostic kits. Protein concentration was determined following the Bradford method using bovine serum albumin as the standard. Each assay was performed in triplicate. The FEUA was then calculated as follows to assess the uricosuric effect of EAF (Perez-Ruiz et al., 2002): FEUA = ([Uur] × [Scr])/([Ucr] × [Sur]) × 100, expressed as a percentage.

2.8.4. Renal histological analyses

Rat's kidneys were fixed for one day at room temperature in fixative (ethanol:chloroform:acetic acid = 6:3:1) and preserved in 70% ethanol. Renal biopsies were dehydrated with a graded series of alcohol and embedded in paraffin. Specimens were cut in 3-μm-thick sections on a rotary microtome and stained with Periodic acid–Schiff reagent for histopathologic evaluation. Renal sections were observed under the light microscope at a magnification of 200×.

2.9. Statistical analysis

Statistical analyses and the IC_{50} values of the samples were calculated with Origin version 6.1 software. All values were expressed as the mean ± S.D. and were analyzed by one-way analysis of variance (ANOVA) and two-tailed Student's *t*-test using SPSS version 13.0 software; a *p*-value of less than 0.05 was considered significant.

3. Results

3.1. Anti-hyperuricemic effect of Smilax china L. fractions in potassium oxonate-induced hyperuricemic mice

Uricase inhibitor potassium oxonate treatment caused hyperuricemia in mice, as indicated with the drastic increases of Sur levels. The oral pretreatment of allopurinol (positive control) at 10 mg/kg dose elicited significantly ($p < 0.001$) reduction of Sur levels in the hyperuricemic mice to the normal value. As shown in Fig. 1, EAF produced a remarkable decrease ($p < 0.01$) in Sur level in oxonate-treated mice after three-time pretreatment orally. There was no significant difference in efficacy between EAF and allopurinol groups. PEF, CF, NF and REF showed no effect on Sur in oxonate-treated mice.

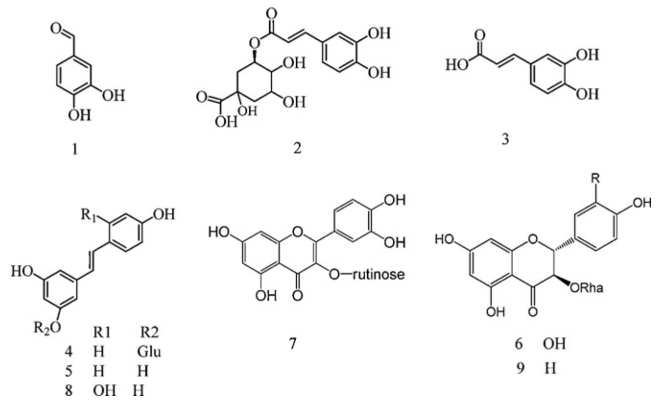

Fig. 2. Structures of the nine compounds identified from EAF of *Smilax china* L. (1) Protocatechuic aldehyde; (2) chlorogenic acid; (3) caffeic acid; (4) polydatin; (5) resveratrol; (6) astilbin; (7) rutin; (8) oxyresveratrol; (9) engeletin.

3.2. Compounds isolated from EAF

Nine compounds, namely protocatechuic aldehyde (1, yield 50 mg), chlorogenic acid (2, yield 203 mg), caffeic acid (3, yield 37 mg), polydatin (4, yield 41 mg), resveratrol (5, yield 152 mg), astilbin (6, yield 187 mg), rutin (7, yield 26 mg), oxyresveratrol (8, yield 55 mg) and engeletin (9, yield 17 mg), were isolated and purified from EAF (200 g). The spectra data were in consistent with the reports in the literatures (Ruan et al., 2005; Xu et al., 2008). Structures of the nine compounds were shown in Fig. 2. Their purities were all proved to be larger than 95% by HPLC analysis.

3.3. HPLC-fingerprint of EAF

The typical HPLC-fingerprint of EAF was shown in Fig. 3. By comparing both the retention times and the UV spectra of the reference standards, nine compounds (protocatechuic aldehyde (1), chlorogenic acid (2), caffeic acid (3), polydatin (4), resveratrol (5), astilbin (6), rutin (7), oxyresveratrol (8) and engeletin (9)) in EAF were well identified.

3.4. Effects of nine compounds on XOD activity and inhibitory modes of action in vitro

3.4.1. Effect on XOD activity

The assays were conducted to investigate whether various concentrations of compounds isolated from EAF inhibited the catalytic activity of XOD. Allopurinol was used as positive control. Results showed that the IC_{50} values of the compound 1, 2, 3, 5, 7 and 8 exhibited strong XOD inhibitory activity *in vitro*, with 50.16, 56.21, 42.60, 37.53, 42.20 and 40.69 μM, respectively (Table 1). Compounds 4, 6 and 9 exhibited low inhibition against the activities of XOD with IC_{50} values of 70.32, 86.27 and more than 100 μM, respectively. Resveratrol (compound 5) with the minimum IC_{50} value inhibited XOD in a concentration-dependent manner among the nine compounds. In addition, the IC_{50} value of allopurinol, a clinical XOD inhibitory drug, was 6.93 ± 0.82 μM under the assay condition and showed the strongest inhibitory activity on XOD.

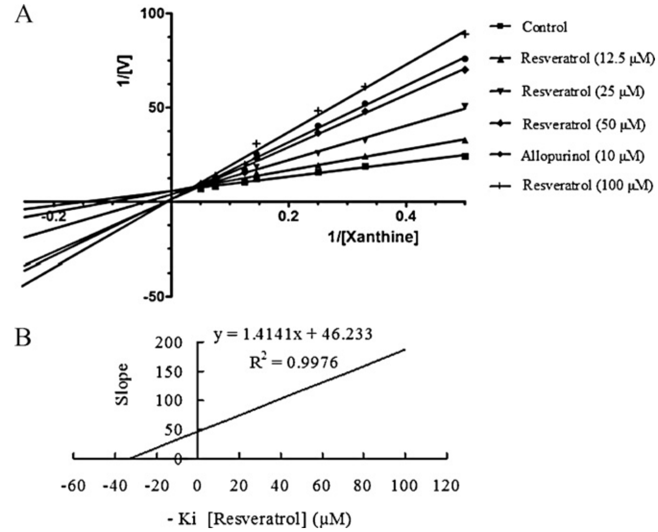

Fig. 4. Inhibition kinetics of XOD by resveratrol. A: Lineweaver–Burk plots in the absence of resveratrol or at different concentrations of resveratrol; B: plots to calculate the inhibition constant K_i. Each datum point was the average value from two determinations in a representative experiment.

3.4.2. Inhibition kinetic analysis

Enzyme inhibition kinetic experiments were carried out to further characterize the inhibitory activities of XOD enzymes by compounds in EAF. Since compound 9 had only weak inhibition on XOD enzymes ($IC_{50} > 100$ μM), it was not investigated further. Based on the analysis of nonlinear regression of inhibition data and Lineweaver–Burk plots, compound 2, 3, 4, 5, 7 and 8 exhibited competitive inhibition against XOD enzyme activities with K_i values of 44, 38, 53, 30, 40 and 33 μM; compound 1 and 6 exhibited mixed type inhibition with K_i values of 42 and 54 μM (as shown in Table 1). The plots to calculate K_i values of resveratrol with strong inhibitory effects was shown in Fig. 4B, and Lineweaver–Burk plots in the absence or presence of resveratrol was shown in Fig. 4A.

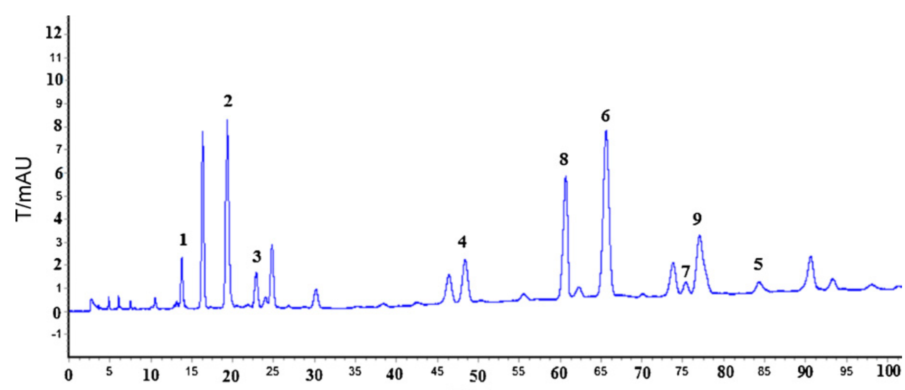

Fig. 3. HPLC chromatogram of EAF. (1) Protocatechuic aldehyde; (2) chlorogenic acid; (3) caffeic acid; (4) polydatin; (5) resveratrol; (6) astilbin; (7) rutin; (8) oxyresveratrol; (9) engeletin.

Table 1
IC_{50} and K_i values of ten tested samples on XOD activity ($n = 3$).

Ingredient	IC_{50} (μM)[a]	K_i (μM)	Mode of inhibition
Protocatechuic aldehyde (1)	50.16 ± 2.28	42	Mixed
Chlorogenic acid (2)	56.21 ± 2.67	44	Competitive
Caffeic acid (3)	42.60 ± 1.42	38	Competitive
Polydatin (4)	70.32 ± 4.53	53	Competitive
Resveratrol (5)	37.53 ± 1.64	30	Competitive
Astilbin (6)	86.27 ± 5.33	54	Mixed
Rutin (7)	42.20 ± 1.74	40	Competitive
Oxyresveratrol (8)	40.69 ± 1.46	33	Competitive
Engeletin (9)	>100	–	–
Allopurinol	6.93 ± 0.82	7.6	Competitive

[a] Data were presented as mean ± S.D. of three determinations.

3.5. Effect of EAF on hyperuricemia and renal dysfunction

3.5.1. Reversed levels of uric acid, creatinine and BUN in fructose-induced hyperuricemic rats

Table 2 summarized the anti-hyperuricemic effect of EAF in hyperuricemic rats. Levels of Sur, Scr and BUN in 10% fructose treated rats were significantly ($p < 0.001$, $p < 0.01$, and $p < 0.001$) increased compared with control rats. EAF at doses of 500, 250 and 125 mg/kg significantly lowered levels of Sur ($p < 0.001$, $p < 0.01$ and $p < 0.05$), Scr ($p < 0.001$, $p < 0.01$, and $p < 0.05$) and BUN ($p < 0.01$, $p < 0.01$, and $p < 0.05$) in hyperuricemic rats to the normal value, and allopurinol at 4 mg/kg also markedly reduced the levels ($p < 0.001$, $p < 0.05$, and $p < 0.05$). Moreover, fructose could induce an obvious ($p < 0.01$ and $p < 0.001$) reduction in Uur and Ucr levels in rats in comparison with control rats. EAF at 500, 250 and 125 mg/kg could significantly elevate Uur ($p < 0.001$, $p < 0.01$ and $p < 0.05$) and Ucr ($p < 0.001$, $p < 0.01$ and $p < 0.05$) levels in hyperuricemic rats, and allopurinol exhibited weaker effect on increasing that ($p < 0.05$ and $p < 0.05$).

FEUA as an important renal uric acid handling parameter was significantly reduced in fructose-fed rats ($p < 0.05$) as previous report (Hu et al., 2009). In this study, FEUA was remarkably reversed by EAF at doses of 500, 250 and 125 mg/kg ($p < 0.01$, $p < 0.01$ and $p < 0.05$) and allopurinol ($p < 0.01$).

In addition, EAF at doses of 500, 250 and 125 mg/kg could significantly ($p < 0.001$, $p < 0.01$ and $p < 0.05$) increase the urinary volume in fructose-fed rats, but allopurinol failed to alter it ($p > 0.05$).

3.5.2. Inhibitory effect on XOD activity

Ten percent of fructose induced a remarkable elevation of hepatic XOD activity in rats ($p < 0.001$) compared with control rats as shown in Fig. 5. EAF at 500, 250 and 125 mg/kg could significantly ($p < 0.01$, $p < 0.01$ and $p < 0.05$) attenuate XOD activity in fructose-fed rats. Allopurinol, at 4 mg/kg, significantly ($p < 0.001$) suppressed hepatic XOD activity of hyperuricemic rats ever to be lower than that of control rats.

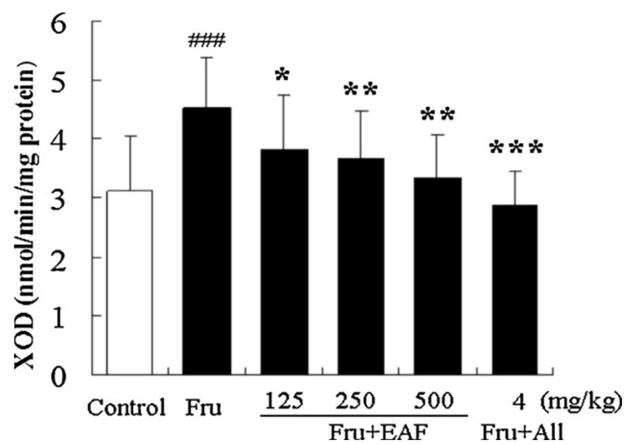

Fig. 5. The inhibitory effects of EAF and allopurinol on XOD activity in liver of fructose-induced hyperuricemic rats (Fru). All: allopurinol. XOD activity in rat's liver were increased by fructose consumption and reversed by EAF (125, 250 and 500 mg/kg) and allopurinol (4 mg/kg). ###$p < 0.001$ versus the control rats; *$p < 0.05$; **$p < 0.01$ and ***$p < 0.001$ versus the fructose-fed rats.

3.5.3. Effect of EAF on improving renal dysfunction

As results aforementioned, levels of Scr, Ucr and BUN could be restored by different doses of EAF in fructose-fed rats. Besides, compared with control rats (Fig. 6A), histological analyses displayed that brush border of epithelial cells was remarkably disappeared and renal tubules were shrank in fructose-fed rats (Fig. 6B). These tubulointerstitial pathologies were ameliorated in some degree after treatment with allopurinol and different doses of EAF (Fig. 6 C–F).

4. Discussion

Hyperuricemia is a main risk factor resulting in gout or chronic nephritis in clinical practice. But currently the therapeutic agents for lowering Sur are sometimes limited due to the associated undesirable adverse effects. A potential source of novel anti-

Table 2
The effects of EAF and allopurinol on serum and urinary levels of uric acid and creatinine, as well as FEUA, BUN and urine volume in fructose-induced hyperuricemic rats.

Group	Dose (mg/kg)	Sur (mg/dL)	Scr (mg/dL)	Uur (mg/dL)	Ucr (mg/dL)	FEUA	BUN (mg/dL)	Urine volume (mL/24 h)
Control	/	2.32 ± 0.15	0.72 ± 0.12	30.99 ± 2.45	122.34 ± 5.56	8.56 ± 1.84	7.45 ± 1.43	11.75 ± 1.75
Fructose	/	3.83 ± 0.24###	1.02 ± 0.15##	18.31 ± 2.08##	79.36 ± 5.24###	6.02 ± 2.42#	11. 32 ± 1.45###	12.43 ± 2.83
	125	3.20 ± 0.28*	0.85 ± 0.09*	24.75 ± 2.31*	85.93 ± 6.23*	7.84 ± 1. 84*	9.03 ± 0.99*	17.33 ± 2.48*
Fructose + EAF	250	2.67 ± 0.13**	0.87 ± 0.10**	26.34 ± 3.52**	92.25 ± 5.36**	9.85 ± 1.57**	8.26 ± 1.78**	19.56 ± 2.78**
	500	2.32 ± 0.15***	0.80 ± 0.08***	32.56 ± 3.45***	100.54 ± 5.36***	11.52 ± 1.03**	8.13 ± 0.23**	23.32 ± 3.43***
Fructose + All	4	2.15 ± 0.21***	0.82 ± 0.12*	20.74 ± 1.78*	86.52 ± 6.45*	9.60 ± 2.56**	9.78 ± 1.24*	15.70 ± 2.19

Data were represented as mean ± S.E.M., 8 rats for each group. Statistical analyses were performed by Student's *t*-test. #$p < 0.05$; ##$p < 0.01$, ###$p < 0.001$ versus the control rats; *$p < 0.05$; **$p < 0.01$ and ***$p < 0.001$ versus the fructose-fed rats.

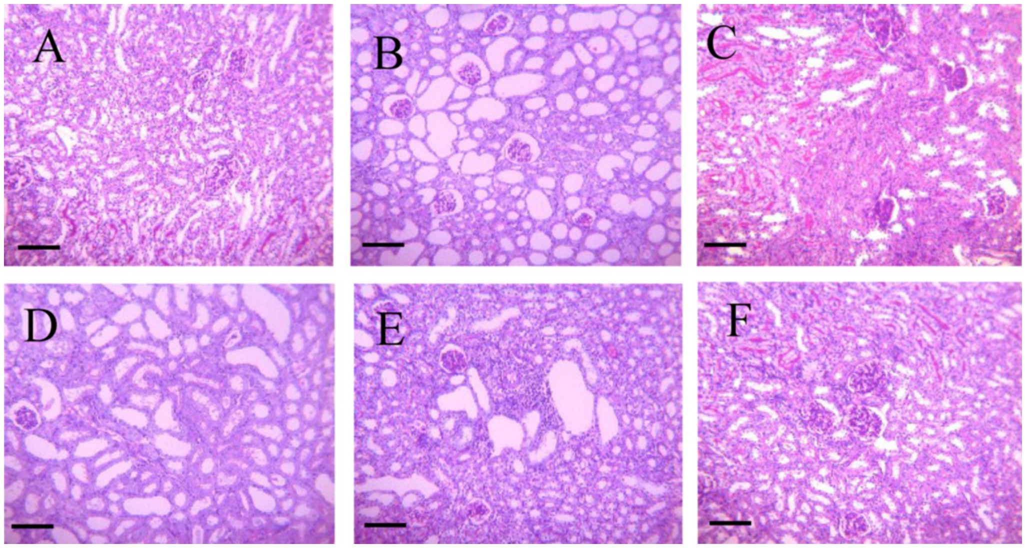

Fig. 6. Representative micrographs of kidney tissue stained with Periodic acid–Schiff from control rats (A), fructose-fed rats (B), fructose + allopurinol (C), fructose + EAF 125 mg/kg (D), fructose + EAF 250 mg/kg (E) and fructose + EAF 500 mg/kg (F). Scale bar: 200 μm; original magnification 200× (A–F).

hyperuricemic agents may be derived from the natural products (Kong et al., 2002). Potassium oxonate is most frequently employed to develop an animal model of hyperuricemia by blocking uricase (Kong et al., 2002). This model is more suitable for preliminary screening of drugs, because it requires a short time and cost less. Hence, anti-hyperuricemic activity of five fractions of *Smilax china* L. rhizome was assessed by assaying Sur in potassium oxonate-induced hyperuricemic mice in our study. The results showed that the fraction of EAF significantly reversed the increase of Sur in potassium oxonate-treated mice (Fig. 1).

Nine compounds were following isolated from EAF through various separation methods and then purified in order to clarify the active component (Fig. 2). The XOD inhibitory activities of the compounds were assayed *in vitro*. The inhibitory activity of nine compounds against XOD from the strongest through weakest was resveratrol, oxyresveratrol, rutin, caffeic acid, protocatechuic aldehyde, chlorogenic acid, polydatin, astilbin, engeletin (Table 1). All of the compounds except Engeletin (IC_{50} > 100 μM) possess certain XOD inhibitory activity. As reported, flavonoid with the hydroxyl groups at C-5 and C-7 and the double bond between C-2 and C-3 were essential for a high inhibitory activity on XOD (Cos et al., 1998), chlorogenic acid showed a weaker activity than caffeic acid (Chan et al., 1995), and resveratrol as one of stilbenes was a XOD inhibitor with IC_{50} 30 μM (Zhou et al., 1999), these results partly supported our study. Moreover, resveratrol, oxyresveratrol, rutin, caffeic acid, chlorogenic acid, polydatin and engeletin, other than protocatechuic aldehyde and astilbin, exerted a competitive inhibition to XOD. Therefore, the results indicate that the XOD inhibition of these compounds contributes to the anti-hyperuricemic effect of EAF.

Increasing evidence has pointed that the worldwide epidemic of metabolic syndrome correlates with an elevation in Sur as well as marked increase in total fructose intake (Nakagawa et al., 2006). Fructose can evaluate Sur that plays an important pathogenetic role in the process of hyperuricemia, and excessive production of uric acid after a load of fructose is related to increased activity of XOD (Fields et al., 1996). Fructose-induced hyperuricemic animal model was applied because it mimics the clinical pharmacology (Nakagawa et al., 2005; Hu et al., 2009). Fructose consumption caused the reduction in Uur and Ucr levels, elevation in Sur, Scr and BUN levels, up-regulation of XOD activity, and renal dysfunction in rats. These data suggest that fructose results in urate under-excretion with impaired renal dysfunction in rats.

Oral administration of EAF significantly reduced Sur level, and down-regulated hepatic XOD in fructose-induced hyperuricemic rats. EAF at the dosage of 125–500 mg/kg exerted inhibitory effect on XOD activity in hyperuricemic rats (Fig. 5) might be one of the anti-hyperuricemic mechanisms. Furthermore, it was interesting to find out that EAF significantly increased the urine volume and FEUA, which suggests that the potent uricosuric effect of EAF might be another anti-hyperuricemic mechanism. However, further researches should be done to study which component of EAF is responsible for anti-hyperuricemic activity, and whether component increase the excretion of uric acid through the regulation of specific transporter molecules such as urate transporter 1 (URAT1) and glucose transporter 9 (GLUT9).

In addition, significantly increasing levels of Scr and BUN indicated the renal dysfunction in fructose-fed rats. EAF markedly reversed the elevations of Scr and BUN to the normal. Moreover, EAF effectively prevented interstitial and tubular damages in the renal histopathologic sections of fructose-induced hyperuricemic rats. As we know, the increase in urinary volume and FEUA helps not only to remove monosodium urate that deposited in renal tissues and then resulted in impaired renal function, but also prevent these crystals growing or aggregating (Arafat et al., 2008). Therefore, the results indicate that the increase in urinary volume and FEUA contribute to the renal protection effect of EAF. So the detailed mechanism of EAF of improving renal function becomes essential to further elucidate in our following research.

5. Conclusion

In conclusion, the results indicate that the anti-hyperuricemic activity of EAF of *Smilax china* L. attributes to the reduction of uric acid production by inhibiting XOD activity and the enhancement of urate excretion by increasing urinary volume and FEUA. Moreover, nine compounds isolated from EAF of *Smilax china* L. showed different XOD inhibitory activities *in vitro* may contribute to the anti-hyperuricemic effect of EAF. These findings provide evidences of anti-hyperuricemic and nephroprotective activities of *Smilax china* L. as a Traditional Chinese Medicine in the treatment of gout.

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References

Anzai, N., Enomoto, A., Endou, H., 2005. Renal urate handling: clinical relevance of recent advances. Current Rheumatology Reports 7, 227–234.

Arafat, O.M., Tham, S.Y., Sadikun, A., Zhari, I., Haughton, P.J., Asmawi, M.Z., 2008. Studies on diuretic and hypouricemic effects of *Orthosiphon stamineus* methanol extracts in rats. Journal of Ethnopharmacology 118, 354–360.

Caulfield, M.J., Munroe, P.B., O'Neill, D., Witkowska, K., Charchar, F.J., Doblado, M., Evans, S., Eyheramendy, S., Onipinla, A., Howard, P., Shaw-Hawkins, S., Dobson, R.J., Wallace, C., Newhouse, S.J., Brown, M., Connell, J.M., Dominiczak, A., Farrall, M., Lathrop, G.M., Samani, N.J., Kumari, M., Marmot, M., Brunner, E., Chambers, J., Elliott, P., Kooner, J., Laan, M., Org, E., Veldre, G., Viigimaa, M., Cappuccio, F.P., Ji, C., Iacone, R., Strazzullo, P., Moley, K.H., Cheeseman, C., 2008. SLC2A9 is a high-capacity urate transporter in humans. PLoS Medicine 5, 197.

Chan, W.S., Wen, P.C., Chiang, H.C., 1995. Structure–activity relationship of caffeic acid analogues on xanthine oxidase inhibition. Anticancer Research 15, 703–707.

Chen, Y., Wang, Q., Li, B., Li, L., Pan, L.N., Huang, Y., 2008. Study on identification and quality for *Smilax China* L. of compound Gout granules. Asia-Pacific Traditional Medicine 4, 237–239.

Cirillo, P., Sato, W., Reungjui, S., Heinig, M., Gersch, M., Sautin, Y., Nakagawa, T., Johnson, R.J., 2006. Uric acid, the metabolic syndrome, and renal disease. Journal of the American Society of Nephrology 17, 165–168.

Cos, P, Ying, L., Calomme, M., Hu, J.P., Cimanga, K., Van Poel, B., Pieters, L., Vlietinck, A.J., Vanden Berghe, D., 1998. Structure–activity relationship and classification of flavanoids as inhibitors of xanthine oxidase and superoxide scavangers. Journal of Natural Products 61, 71–76.

Fields, M., Lewis, C.G., Lure, M.D., 1996. Allopurinol, an inhibitor of xanthine oxidase, reduces uric acid levels and modifies the signs associated with copper deficiency in rats fed fructose. Free Radical Biology & Medicine 20, 595–600.

Hammer, B., Link, A., Wagner, A., Bohm, M., 2001. Fatal allopurinol-induced hypersensitivity in asymptomatic hyperuricemia syndrome. Deutsche Medizinische Wochenschrift 126, 1331–1334.

Horiuchi, H., Ota, M., Nishimura, S., Kaneko, H., Kasahara, Y., Ohta, T., Komoriya, K., 2000. Allopurinol induces renal toxicity by impairing pyrimidine metabolism in mice. Life Science 66, 2051–2070.

Hu, Q.H., Wang, C., Li, J.M., Zhang, D.M., Kong, L.D., 2009. Allopurinol, rutin, and quercetin attenuate hyperuricemia and renal dysfunction in rats induced by fructose intake: renal organic ion transporter involvement. American Journal of Physiology: Renal Physiology 297, 1080–1091.

Iseki, K., Oshiro, S., Tozawa, M., Iseki, C., Ikemiya, Y., Takishita, S., 2001. Significance of hyperuricemia on the early detection of renal failure in a cohort of screened subjects. Hypertension Research 24, 691–697.

Iseki, K., Ikemiya, Y., Inoue, T., Iseki, C., Kinjo, K., Takishita, S., 2004. Significance of hyperuricemia as a risk factor for developing ESRD in a screened cohort. American Journal of Kidney Diseases 44, 642–650.

Ishizaka, N., Ishizaka, Y., Toda, E., Nagai, R., Yamakado, M., 2005. Association between serum uric acid, metabolic syndrome, and carotid atherosclerosis in Japanese individuals. Arteriosclerosis, Thrombosis, and Vascular Biology 25, 1038–1044.

Kong, L.D., Zhou, J., Wen, Y.L., Li, J.M., Cheng, C.H.K., 2002. Aesculin possesses potenthy pouricemicaction in rodents but is devoid of xanthine oxidase dehydrogenase inhibitory activity. Planta Medica 68, 175–178.

Miatello, R., Vázquez, M., Renna, N., Cruzado, M., Zumino, A.P., Risler, N., 2005. Chronic administration of resveratrol prevents biochemical cardiovascular changes in fructose-fed rats. American Journal of Hypertension 18, 864–870.

Nakagawa, T., Tuttle, K.R., Short, R.A., Johnson, R.J., 2005. Hypothesis: fructose-induced hyperuricemia as a causal mechanism for the epidemic of the metabolic syndrome. Nature Clinical Practice Nephrology 1, 80–86.

Nakagawa, T., Hu, H., Zharikov, S., Tuttle, K.R., Short, R.A., Glushakova, O., Ouyang, X., Feig, D.I., Block, E.R., Herrera-Acosta, J., Patel, J.M., Johnson, R.J., 2006. A causal role for uric acid in fructose-induced metabolic syndrome. American Journal of Physiology: Renal Physiology 290, F 625–F 631.

Perez-Ruiz, F., Calabozo, M., Erauskin, G.G., Ruibal, A., Herrero-Beites, A.M., 2002. Renal underexcretion of uric acid is present in patients with apparent high urinary uric acid output. Arthritis & Rheumatism 47, 610–613.

Ruan, H.L., Zhang, Y.H., Zhao, W., Tan, Y.F., Sun, Z.L., Wu, J.Z., 2002. Studies on the chemical constituents of *Smilax china* L. Natural Product Research and Development 14, 35–36.

Ruan, J.L., Zou, J.H., Cai, Y.L., 2005. Studies on chemical constituents of *Smilax china*. Journal of Chinese Medicinal Materials 28, 24–26.

Schlesinger, N., 2004. Management of acute and chronic gouty arthritis: present state-of-the-art. Drugs 64, 2399–2416.

Shu, X.S., Gao, Z.H., Yang, X.L., 2006. Anti-inflammatory and anti-nociceptive activities of *Smilax china* L. aqueous extract. Journal of Ethnopharmacology 103, 327–332.

Stavric, B., Clayman, S., Gradd, R.E., Hebert, D., 1975. Some in vivo effects in the rat induced by chlorprothixene and potassium oxonate. Pharmacology Research Communication 7, 117–124.

State Administration of Traditional Chinese Medicine of People's Republic of China (Ed.), 1999. Zhong-hua-ben-cao. Shanghai Science and Technology Publisher, Shanghai, China, pp. 157–160.

State Pharmacopoeia Commission of People's Republic of China (Ed.), 2010. Pharmacopoeia of People's Republic of China. China Medical Science and Technology Publishing House, Beijing, China, p. 290.

Terkeltaub, R.A., 2003. Clinical practice. Gout. The New England Journal of Medicine 349, 1647–1655.

Weiner, D.E., Tighiouart, H., Elsayed, E.F., Griffith, J.L., Salem, D.N., Levey, A.S., 2008. Uric acid and incident kidney disease in the community. Journal of the American Society of Nephrology 19, 1204–1211.

Wortmann, R.L., 2002. Gout and hyperuricemia. Current Opinion in Rheumatology 14, 281–286.

Wu, L.S., Wang, X.J., Wang, H., Yang, H.W., Jia, A.Q., Ding, Q., 2010. Cytotoxic polyphenols against breast tumor cell in *Smilax china* L. Journal of Ethnopharmacology 130, 460–464.

Xu, Y., Liang, J.Y., Zou, Z.M., 2008. Studies on chemical constituents of rhizomes of *Smilax china* L. China Journal of Chinese Materia Medica 21, 119–121.

Yoo, T.W., Sung, K.C., Shin, H.S., Kim, B.J., Kim, B.S., Kang, J.H., Lee, M.H., Park, J.R., Kim, H., Rhee, E.J., Lee, W.Y., Kim, S.W., Ryu, S.H., Keum, D.G., 2005. Relationship between serum uric acid concentration and insulin resistance and metabolic syndrome. Circulation Journal 69, 928–933.

Yu, Z.F., Fong, W.P., Cheng, Christopher, H.K., 2006. The dual actions of morin (3,5,7,2′,4′-Pentahydroxyflavone) as a hypouricemic agent: uricosuric effect and xanthine oxidase inhibitory activity. Journal of Pharmacology and Experimental Therapeutics 316, 169–175.

Zhou, C.X., Tanaka, J., Cheng, C.H., Higa, T., Tan, R.X., 1999. Steroidal alkaloids and stilbenoids from *Veratrum taliense*. Planta Medica 65, 480–482.

Zhou, X., Matavelli, L., Frohlich, E.D., 2006. Uric acid: its relationship to renal hemodynamics and the renal renin–angiotensin system. Current Hypertension Reports 8, 120–124.

Zhu, J.X., Wang, Y., Kong, L.D., Yang, C., Zhang, X., Ji, X.Z., Ying, W., Ling, D.K., Cheng, Y., Xin, Z., 2004. Effects of Biota orientalis extract and its flavonoid constituents, quercetin and rutin on serum uric acid levels in oxonate-induced mice and xanthine dehydrogenase and xanthine oxidase activities in mouse liver. Journal of Ethnopharmacology 93, 133–140.